

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Develop a Styled Webpage Using HTML5 Canvas and CSS3

2. AIM

To design and develop a styled webpage as a static web page using HTML5 and CSS3, demonstrating HTML5 Canvas graphics, gradient backgrounds, box and text shadows, rounded corners, Flexbox/Grid layout, and CSS3 transitions with hover effects.

3. OBJECTIVE

The objectives of this experiment are:

- To understand and use the HTML5 <canvas> element for rendering graphics.
 - To draw basic shapes — a rectangle, a circle (arc), and a line — using the Canvas 2D drawing API.
 - To render text on a canvas using the fillText() method with a custom font.
 - To apply CSS3 background gradients using linear-gradient().
 - To apply box-shadow and text-shadow for depth and visual emphasis.
 - To use border-radius for rounded corners on containers, cards, and buttons.
 - To lay out page sections using CSS3 Flexbox and CSS Grid (grid-template-columns, auto-fit).
 - To implement smooth CSS3 transitions and hover effects on navigation links, buttons, and cards.
 - To structure the page using semantic HTML5 elements: header, nav, main, section, article, footer.
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4. THEORY

HTML5 Canvas is a rectangular drawing surface exposed through the <canvas> element. JavaScript accesses a 2D rendering context via getContext('2d') and issues drawing commands —

fillRect()/strokeRect() for rectangles, arc() for circles, moveTo()/lineTo() for lines, and fillText() for text — to render graphics dynamically in the browser.

CSS3 Gradients — linear-gradient() blends two or more colours along a defined angle, producing smooth background transitions without image files.

Box Shadow & Text Shadow — box-shadow adds layered shadows around an element's box, while text-shadow adds shadow behind text; both accept offset, blur, and colour (including rgba transparency) to create depth.

Border Radius — border-radius rounds the corners of an element's box, used here on containers, cards, and buttons for a softer, modern look.

Flexbox & CSS Grid — Flexbox (display: flex) arranges items along a single axis, used for the page's overall vertical layout and the navigation bar; CSS Grid (display: grid, grid-template-columns with repeat/auto-fit) arranges the content cards in a responsive, self-wrapping layout.

CSS3 Transitions — the transition property animates changes to properties such as background-color, transform, and box-shadow smoothly over a set duration, producing the hover effects on links, buttons, and cards.

5. ALGORITHM / PROCEDURE

- Create an HTML5 file named My_Styled_Webpage.html.
 - Add the <!DOCTYPE html> declaration and set the language attribute on <html>.
 - Inside <head>, include the charset meta tag, viewport meta tag, page title, and an internal <style> block.
 - In the <style> block, reset default margin/padding with a universal selector and set box-sizing: border-box.
 - Style the <body> with a font stack, line-height, text colour, and a linear-gradient background; set display: flex with flex-direction: column.
 - Build the <header> with a background colour, padding, box-shadow, and a text-shadow heading.
 - Build the <nav> with a flex-based, centred link list and a hover transition on each link.
 - Add a <main> element to hold the primary content, constrained with max-width and centred margins.
 - Create a canvas <section> with a rounded, shadowed white container holding a <canvas> element.
 - Write a <script> block to get the 2D context and draw: one filled/stroked rectangle, one filled/stroked circle (arc), one line, and one text string with fillText().
 - Create a cards <section> using CSS Grid (auto-fit, minmax) holding three <article class="card"> elements.
 - Style each .card with padding, border-radius, box-shadow, and a hover transform/shadow transition.
 - Add a styled .btn button inside each card with border-radius, box-shadow, and a hover colour/scale transition.
 - Add a <footer> with a background colour, top border accent, and a copyright notice.
 - Open the file in a web browser and verify the canvas drawing and all CSS3 effects.
 - Record expected vs. actual output for each test case.
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6. PROGRAM

My_Styled_Webpage.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }
    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      line-height: 1.6;
      color: #333;
      background: linear-gradient(135deg, #e0c3fc 0%, #8ec5fc 100%);
      min-height: 100vh;
      display: flex;
      flex-direction: column;
    }
    header {
      background-color: #2c3e50;
      color: #ffffff;
      padding: 2rem 1rem;
      text-align: center;
      box-shadow: 0 4px 6px rgba(0, 0, 0, 0.15);
    }
    header h1 {
      font-size: 2.5rem;
      text-shadow: 2px 2px 4px rgba(0, 0, 0, 0.5);
```

```
margin-bottom: 0.5rem;
}
header p { font-size: 1.1rem; color: #ecf0f1; }
nav {
  background-color: #34495e;
  box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
}
nav ul { display: flex; justify-content: center; list-style: none; }
nav ul li a {
  display: block;
  color: #ffffff;
  text-decoration: none;
  padding: 1rem 1.5rem;
  font-weight: 600;
  transition: background-color 0.3s ease, color 0.3s ease;
}
nav ul li a:hover { background-color: #1abc9c; color: #ffffff; }
main {
  flex: 1;
  max-width: 1100px;
  margin: 2rem auto;
  padding: 0 1rem;
  width: 100%;
}
.canvas-section {
  background: #ffffff;
  padding: 1.5rem;
  margin-bottom: 2rem;
  border-radius: 12px;
  border: 1px solid #dcdde1;
  box-shadow: 0 8px 16px rgba(0, 0, 0, 0.1);
  text-align: center;
}
.canvas-section h2 {
```

```
margin-bottom: 1rem;
font-family: 'Georgia', serif;
color: #2c3e50;
}
#myCanvas {
background-color: #fafafa;
border: 2px dashed #3498db;
border-radius: 8px;
max-width: 100%;
height: auto;
}
.cards-container {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
gap: 1.5rem;
}
.card {
background: #ffffff;
padding: 1.5rem;
border-radius: 12px;
border: 1px solid #e0e0e0;
box-shadow: 0 4px 8px rgba(0, 0, 0, 0.08);
display: flex;
flex-direction: column;
justify-content: space-between;
transition: transform 0.3s ease, box-shadow 0.3s ease;
}
.card:hover {
transform: translateY(-5px);
box-shadow: 0 8px 20px rgba(0, 0, 0, 0.15);
}
.card h3 {
font-family: 'Arial', sans-serif;
font-size: 1.3rem;
```

```
    color: #2c3e50;
    margin-bottom: 0.75rem;
    text-shadow: 1px 1px 2px rgba(0, 0, 0, 0.1);
}
.card p { font-size: 0.95rem; color: #555; margin-bottom: 1.5rem; }
.btn {
    align-self: start;
    padding: 0.6rem 1.2rem;
    background-color: #3498db;
    color: white;
    border: none;
    border-radius: 20px;
    font-size: 0.9rem;
    font-weight: bold;
    cursor: pointer;
    box-shadow: 0 2px 5px rgba(0, 0, 0, 0.2);
    transition: background-color 0.3s ease, transform 0.2s ease;
}
.btn:hover { background-color: #2980b9; transform: scale(1.05); }
footer {
    background-color: #2c3e50;
    color: #bdc3c7;
    text-align: center;
    padding: 1.5rem;
    margin-top: auto;
    font-size: 0.9rem;
    border-top: 3px solid #1abc9c;
}
</style>
</head>
<body>

<header>
<h1>My Styled Webpage</h1>
<p>Exploring HTML5 Canvas & CSS3 Styling Concepts</p>
```

```
</header>
```

```
<nav>
```

```
<ul>
```

```
<li><a href="#canvas">Canvas Graphic</a></li>
```

```
<li><a href="#cards">Content Cards</a></li>
```

```
<li><a href="#about">About</a></li>
```

```
</ul>
```

```
</nav>
```

```
<main>
```

```
<section class="canvas-section" id="canvas">
```

```
<h2>HTML5 Canvas Graphics</h2>
```

```
<canvas id="myCanvas" width="500" height="250">
```

```
  Your browser does not support the HTML5 canvas tag.
```

```
</canvas>
```

```
</section>
```

```
<section class="cards-container" id="cards">
```

```
<article class="card">
```

```
<h3>CSS3 Typography & Color</h3>
```

```
<p>Demonstrating custom font families, controlled text sizing, text shadows, and subtle foreground/background contrast rules for improved UI visual hierarchy.</p>
```

```
<button class="btn">Read Article</button>
```

```
</article>
```

```
<article class="card">
```

```
<h3>Box Model Mastery</h3>
```

```
<p>Showcasing margins, padding, custom borders, rounded corners (border-radius), and layered box-shadows to build clean, modern content containers.</p>
```

```
<button class="btn">Learn More</button>
```

```
</article>
```

```
<article class="card">
```

```
<h3>Interactive Transitions</h3>
```

```
<p>Enhancing user experience with CSS3 hover effects on buttons and dynamic layout elements like navigation links and elevated cards.</p>
```

```
<button class="btn">Explore UI</button>
</article>
</section>
</main>

<footer id="about">
  <p>&copy; 2026 My Styled Webpage</p>
</footer>

<script>
  const canvas = document.getElementById('myCanvas');
  const ctx = canvas.getContext('2d');

  ctx.fillStyle = '#3498db';
  ctx.fillRect(30, 40, 120, 80);
  ctx.strokeStyle = '#2c3e50';
  ctx.lineWidth = 3;
  ctx.strokeRect(30, 40, 120, 80);

  ctx.beginPath();
  ctx.arc(250, 80, 40, 0, 2 * Math.PI);
  ctx.fillStyle = '#e74c3c';
  ctx.fill();
  ctx.lineWidth = 3;
  ctx.strokeStyle = '#c0392b';
  ctx.stroke();

  ctx.beginPath();
  ctx.moveTo(330, 120);
  ctx.lineTo(460, 40);
  ctx.lineWidth = 4;
  ctx.strokeStyle = '#1abc9c';
  ctx.lineCap = 'round';
  ctx.stroke();
```



```

    ctx.font = 'bold 22px "Segoe UI", sans-serif';
    ctx.fillStyle = '#2c3e50';
    ctx.textAlign = 'center';
    ctx.fillText('Purna Chandra', 250, 200);
</script>

</body>
</html>

```

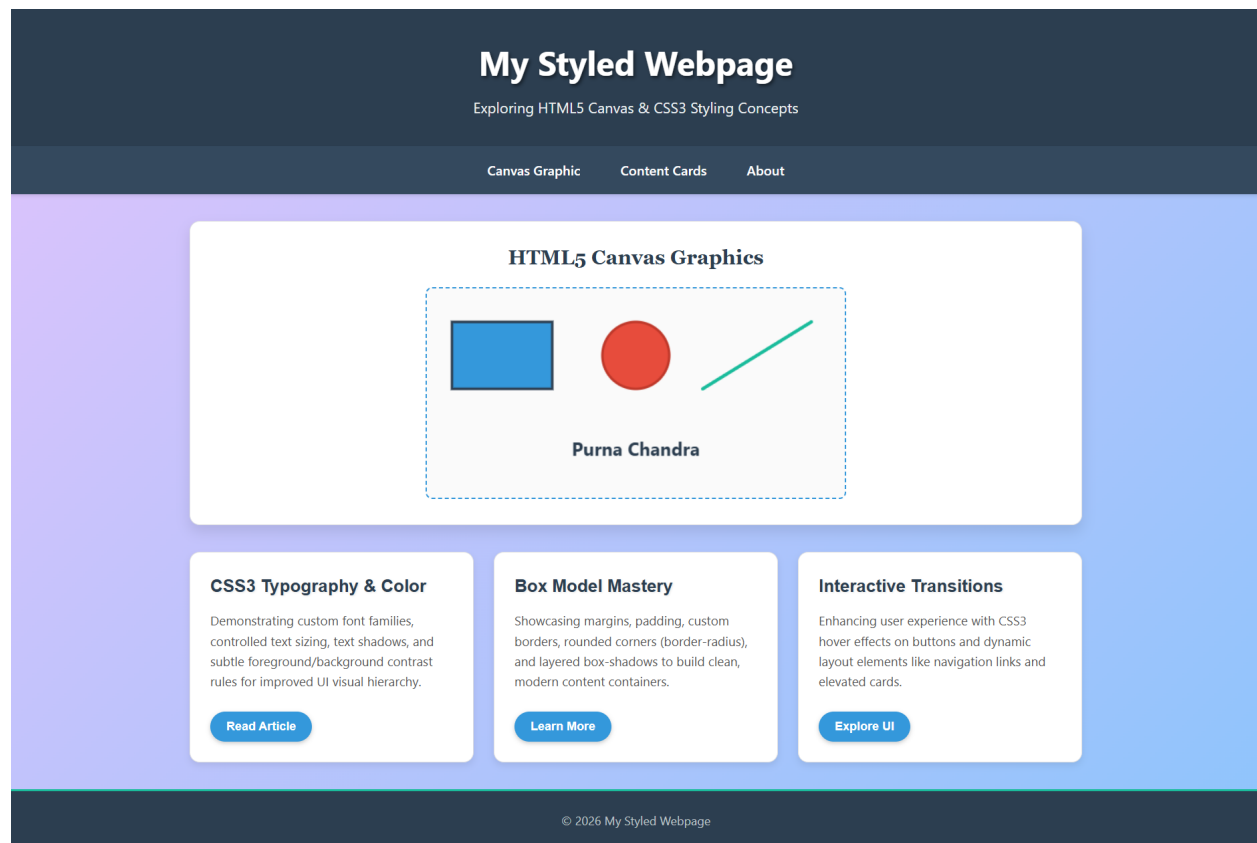
7. CODE EXPLANATION

Code	Explanation
<canvas id="myCanvas">	Defines the drawing surface; width/height set its pixel resolution.
getContext('2d')	Retrieves the 2D rendering context used to issue drawing commands.
fillRect() / strokeRect()	Draws a filled and outlined rectangle on the canvas.
arc()	Draws a circular path, filled and stroked to render the circle.
moveTo() / lineTo()	Defines the start and end points of the drawn line.
fillText()	Renders the student's name as text on the canvas using a custom font.
linear-gradient()	Produces the diagonal purple-to-blue gradient background on the page body.
box-shadow	Adds layered shadows to the header, canvas container, cards, and buttons for depth.
text-shadow	Adds a subtle shadow behind the header heading and card titles.
border-radius	Rounds the corners of the canvas container, cards, and buttons.
display: flex	Arranges the body layout vertically and centres the navigation links.
display: grid	Arranges the three content cards in a responsive, auto-wrapping grid.
transition	Animates hover changes smoothly for links, cards, and buttons.
:hover	Pseudo-class defining the visual state changes on mouse-over.
<header> <nav> <main> <section> <article> <footer>	Semantic HTML5 elements structuring the page content.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Load the page in a browser	Page renders with gradient background and all sections visible	Page loaded correctly; gradient background displayed	Pass
TC-02	Inspect the canvas graphic	Rectangle, circle, line, and name text render inside the canvas	All four shapes/text rendered correctly on canvas	Pass
TC-03	Hover over a navigation link	Link background changes colour with a smooth transition	Background changed to teal smoothly on hover	Pass
TC-04	Hover over a content card	Card lifts up with an enlarged shadow	Card translated upward; shadow enlarged	Pass
TC-05	Hover over a card button	Button darkens and scales slightly	Button colour darkened; scale transform applied	Pass
TC-06	Resize the browser window (narrow width)	Cards re-flow into a single column via CSS Grid auto-fit	Cards stacked into one column below the breakpoint width	Pass
TC-07	Inspect header heading text	Heading displays with a visible drop text-shadow	Text-shadow rendered behind the heading text	Pass
TC-08	Inspect canvas container corners	Container shows rounded corners and a soft box-shadow	Rounded corners and shadow displayed as styled	Pass
TC-09	Click a navigation anchor link	Page scrolls to the corresponding section id	Browser scrolled to the target section	Pass
TC-10	Inspect footer	Footer displays a teal top border and copyright text	Footer rendered with border accent and copyright notice	Pass

9. OUTPUT



10. RESULT

Thus, a styled webpage was successfully designed and developed using HTML5 Canvas and CSS3. The page demonstrates canvas-based graphics rendering, gradient backgrounds, box and text shadows, rounded corners, Flexbox/Grid layout, and smooth CSS3 transitions with hover effects. All ten test cases passed successfully, confirming correct rendering and interactive behaviour across the page.